

LegalEase: AI-Powered Legal Document Generator

Requirement Analysis Phase

1. Team Details

Team Name: LegalEase: AI-Powered Legal Document generator

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2. Introduction

Requirement Analysis is the phase in which the team identifies the needs, functions, and technical requirements of the LegalEase project. This phase helps define what the system should do and what resources are required to develop it.

3. Problem Analysis

Manual preparation of legal documents can be time-consuming and may require knowledge of legal formats. Users need a simple way to provide basic information such as parties, terms, and dates and receive a structured legal document draft.

4. Functional Requirements

- The system should allow users to select a legal document type.
- The system should accept party details.
- The system should accept terms and conditions.
- The system should accept the effective date.
- The system should send the entered information to the backend.
- The system should use Generative AI to create a document draft.
- The system should display the generated document for preview.
- The system should allow the generated document to be edited.
- The system should provide PDF and Word download options.

5. Non-Functional Requirements

- The application should have a simple and user-friendly interface.
- The system should provide reliable API communication.
- Required inputs should be validated.
- The application should provide reasonable response time subject to the AI service.

- API credentials should be stored securely.
- Generated documents should be readable and properly formatted.

6. Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM recommended
- Internet connection
- Keyboard and mouse

7. Software Requirements

- Python
- Visual Studio Code
- FastAPI
- Uvicorn
- Streamlit
- Google Gemini API / Google Gen AI SDK
- python-docx
- FPDF / ReportLab
- Requests
- python-dotenv

8. User Requirements

The user should be able to select a document type and enter the required information without advanced technical knowledge. The user should be able to preview, edit, and download the generated document.

9. System Requirements

LegalEase requires a Streamlit frontend, FastAPI backend, and Gemini-powered AI module. The frontend collects the user inputs, the backend processes the request, and the AI module generates the document draft.

10. Input Requirements

- Document Type
- Parties
- Terms and Conditions
- Effective Date

11. Output Requirements

- Generated legal document draft
- Editable document content
- PDF document
- Word document

12. Security Requirements

The Gemini API key should be stored securely using environment variables and should not be exposed in public source code or uploaded to GitHub. Generated documents should be reviewed by a qualified legal professional before real-world use.

13. Requirement Analysis Conclusion

The requirement analysis identifies the functional, non-functional, hardware, software, user, system, input, output, and security requirements of LegalEase. These requirements provide a clear foundation for the design and development of the project.